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Appendix to Chapter 15

The No-Limit AKQ Game

Summary

The solution to the full no-limit AKO game is surprisingly close to the solution to the spread-limit game from Chapter 17. The steps in deriving the strategy are:

- We can first solve a full-street game where X may not bet kings (or is known not to bet them). We find that X bets *r* (as was the case for Y in the half-street game) with some aces and checks the remaining aces, while Y bets a different, larger amount s_0 if checked to. Both players balance their strategies with aces and queens appropriately.
- X can then improve on his strategy if we relax the constraint on betting kings by betting a smaller amount *t* with some kings.
- Y responds to X betting *t* with kings by raising to another value *s(t)* and raise-bluffing with queens with some frequency β. By doing this he makes X at most indifferent to checking or betting an amount *t* with kings. β is related to s_0, the amount that Y bets when X checks.
- X responds by betting *t* with some aces in addition to kings and gaining value by picking off raise-bluffs and getting called by Y's kings. Y cannot prevent X from gaining value in one of these two ways.
- However, hi the no-limit case. X cannot simply bet all his aces (as he did in the spread-limit case). If he does, and leaves himself with only kings in his checking distribution, Y can exploit him by betting a large amount with aces and some queens. Since he has only kings, he must call sometimes, which is more costly than the amount he can gain by betting aces.
- We can construct a function A(*t*), which is the value of betting *t* with aces. This is the sum of the values of being called by kings and picking off raise-bluffs when the opponent holds queens.
- For any particular s_0 value, we can find the optimal *t* value, by maximizing A(*t*), We call this value τ. This is the amount that X should bet with some aces and some kings.
- Since he will be playing a mixed strategy of betting some aces and checking some aces, he must have the same expectation from checking and betting τ; that is A(0) = A(τ).
- We can then find a raise amount t_2 that Y would like to raise to. Unfortunately, the rules force Y to raise to at minimum 2_t.
- We then balance the strategies as appropriate with bluffs, calls, and raise-bluffs such that each player is indifferent to changing various parts of his strategy.

The final strategies are: (all decimal values are approximate, and all percentages are "of the hands he has played this far in this way")

Solution:

X:

Checks 61.2% of his aces, and if Y bets behind, X raises all-in.
Bets τ = 0.2521 with the rest of his aces, and reraises all-in if Y raises.

Bets τ = 0.2521 with 15.62% of his kings and folds to any raise.
Checks the remainder of his kings and calls a bet of *s* with an appropriate

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fraction of kings such that his total calling or raising frequency with hands he checked is $1/(1 + s)$.

Bets $\tau = 0.2521$ with 7.81% of his queens and folds to any raise.
Checks and folds with the remaining 92.19% of his queens.

Y:

If X checks, Y bets $s_0 = 0.5422$ with all his aces and 35.16% of his queens.
If X check-raises, Y reraises all-in with his aces and folds his queens.
If X bets an amount t, Y responds as follows:

Define the following:
s_0 is Y's bet amount if X checks = 0.5422.
$\alpha_0 = s_0/(1+s_0)$
$t_2 = \alpha_0/(1 - \beta)$ (if this is greater than $2t$), otherwise $2t$
$\beta = (\alpha_0 - t)/(1 + t)$ if $\alpha_0 > t$, 0 otherwise.

Y raises to t_2 with all aces and with β of his queens. If X raises again, Y raises all-in with aces and folds queens.

Y calls with enough kings to make his total calling frequency with hands that can beat a queen $1/(1 + t)$.

Derivation

First we consider the game with the constraint that X may not bet kings. We can see immediately that if he bets with all aces, he can be exploited by Y betting a very large amount with all aces and an appropriate fraction of queens. Since he has bet all aces, he is left with only kings to defend against the opponent's bluffs. He must call with an appropriate fraction of kings, but gives up too much value to the opponents' aces to make this worthwhile.

To see this clearly, presume that X bets all his aces and checks all his kings. Then suppose Y bets 100 units (where pot = 1) with all his aces and 100/101 of his queens. If he does this X must call with $^1/_{101}$ of his kings to prevent Y from bluffing. This means he is giving up $^{100}/_{101}$of a unit when he holds a king and X holds an ace. But he could improve by checking a few ace; and calling the bets which will all be bluffs). So betting all aces initially must not be optimal.

Since he must check all kings (by the rules of the game), and he must check some aces to provide a credible threat against the opponent's bluffing large amounts, he may either check all aces, or simply some fraction of aces. Call the fraction of aces he checks Ak. X will then bet some amount t with the aces he does not check and an appropriate fraction of queens. If he bets an amount t, then Y will respond by (at least) calling with a total fraction of his hands equal to $1/(1 + t)$. This will make X indifferent to bluffing queens. If X checks, then Y will bet an amount s. Thus we have three unknown elements of the strategy, in addition to the various easy-to-calculate betting and bluffing ratios.

For Y, the fraction of all calling hands that are kings is $1/(1 + t) - {}^1/_2$ unless X bets an amount greater than 1, in which case he will call with all aces and no kings. X, then, has the following equity from betting t when he holds an ace and Y holds a king:

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<A vs. K, bet t > = $(t)(2/(1 + t) - 1) = (t - t)^2/(t + 1)$

Maximizing this with respect to t:

$(t + 1)(1 - 2t) - (t - t^2)(1) / (t + 1)^2 = 0$
$1 - t^2 - 2t = 0$
$t = \sqrt{2} - 1$

Recalling Equation 14.1:

$t = r$

Y will then call with $(t - 1)/(t + 1)$ of his kings as described above.

When X checks, we have a similar situation, Y will bet some amount s with his aces and appropriate queens. X will call with the aces that remain (remember, he bet some aces), and then enough kings to make Y indifferent to bluffing with queens. We know that X must be indifferent to betting r or checking with his aces because he will play a mixed strategy of these two options. The expectations of the two possible plays with an ace are:

<A, check > = $(s (1 + s))(s)(1/2) = s^2 / 2 (1+s)$
<A, bet r > = $(r - r^2)/2(1 + r) = 3/2 - \sqrt{2}$

Setting these equal:

$3/2 - \sqrt{2} = s^2 / 2 (1+s)$
$3 - 2\sqrt{2} + 3s - 2\sqrt{2}s = s^2$
$s \approx 0.50879$

The third remaining unknown of the final strategy is the frequency A_k with which X checks. As we saw earlier, if A_k is not chosen properly, then Y can exploit X by bluffing larger or smaller amounts. Hence X will select A_k such that Y's equity is maximized by betting s.

When X holds a king and Y an ace, Y gains one bet (call this bet q) when X calls. X's calling frequency, in turn, depends on A_k. Recall that this value is the fraction of aces that remain. The total hands that X has remaining is $A_k(1/2) + 1/2$, or $1/2 (l + A_k)$. If Y bets s, then X will call with a total fraction of hands (aces and kings) equal to:

$(1/2)(1 + A_k)(1/(1 + q)) = (1 + A_k)/2(1 + q)$

Of these, X will call with aces $1/2\ A_k$ of the time, and then with remaining kings as necessary. The number of kings that X will call with is:

$(1 + A_k)/2(1 + q) - 1/2\ A_k$
$= (l+A_k - (Ak+qA_k))/2(l + q)$
$= (1 - qA_k)/2(l + q)$

and Y's equity from being called by kings is this value times q, or

<K/A, bet q > $=(q - q2A_k)/2(1 + q)$.

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We can maximize this expression for Y:

$(1 + q)\ (1 - 2qA_k) - (q - q^2Ak))/(1 + q)^2 = 0$
$L = 2qA_k + q^2A_k$

Completing the square, we have:

$(q + 1)^2 = 1/A_k + 1$
$q + 1 = \pm\sqrt{(A_k + 1)}/A_k$
$q = \sqrt{(A_k + 1)}/A_k - 1$

Thus, in response to a choice of A_k by X, Y maximizes his equity by betting an amount q equal to this value. By setting $q = s$, we can find the optimal choice of A_k for X.

$0.50879 = \sqrt{(A_k + 1)}/A_k - 1$
$A_k \approx 0.78342$

So the final strategy for this simplified game (where X must check kings) is:

X checks aces about 78.3% of the time, bets r (about 41.2% of the pot) with the remainder, bets r times the pot with $r/(1 + r)$ times as many queens, or about 6.3% of his queens overall, and is forced to check all kings.

Y responds to a bet of t (t > 0) by calling with (at minimum) all aces, and if the bet size is small enough, with enough kings to make his total calling frequency (if the opponent is bluffing) equal to $1/(1 + s)$

If X checks, Y bets s (about 50.8% of the pot) with all aces and about 33.7% of his queens. X then calls with enough kings to make his total calling frequency (with the aces he checked and kings) equal to $1/(1 + s)$. For the optimal bet sizes, these calling frequencies are: Y calls an r sized bet with r of his kings, and X calls an s size bet with about 39.8% of his kings.

The value of this simplified game to Y is approximately 0.02761 units.

If we relax the "must check kings" restriction on the game, however, X can now improve. When X checks a king, we can use a shortcut to find his equity. Since we know that Y will make him indifferent to calling with a king by bluffing and betting in the proper ratio, we can assume he folds all the time. He loses one unit whenever a queen bluffs. Y will bluff an amount $s_0/(1 + s_0)$ of the time and have a queen half the tune. We define $\alpha_0 = s_0(1 + s_0)$. Thus his equity is:

$\langle P1, \text{check K}\rangle = (^1/_2)\ [- s/(1 + s)] = - \alpha_0/2$

Obviously if Y continues to respond to a bet of t by calling with aces and kings, X can bet a tiny amount and prevent Y from bluffing. So let us take as our candidate strategy that Y will respond to a bet of t with a raise to either $2t$ (the minimum allowed) or some other value t_2, whichever is greater, with all aces and with some fraction of queens $\beta(t)$.

Then X's equity of betting t with a king is:

$\langle P1, \text{bet } t/\text{K}\rangle = (^1/_2)(-t) + (^1/_2)(\beta)\ [-(1 + t)]$

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β must be large enough that this expression is smaller than the first expression; that is, that X's equity from betting *t* is less than or equal to his equity from checking. The cases where his equity from betting *t* is less than his equity occur when he bets too much with a king. For now we will concern ourselves only with reasonable preemptive bet sizes. (Later we will see that p can be zero if the bet size is large enough.) So we have:

$(^1/_2)(-t) + (^1/_2)(\beta)\,[-(1 + t)] = -\,\alpha_0/2$
$t + \beta t + \beta = \alpha_0$
$\beta = (\alpha_0 - t)(1+t) \qquad (\alpha_0 > t)$

Recall that when X bets, Y calls with $(1 - t)/(1 + t)$ of his kings. If t_2 is the amount that Y raises to, then the equity of betting *t* with an ace is:

$A(t) = (^1/_2)(t(1 - t)/(1 + t)) + (^1/_2)(\beta)(t_2)$

If we substitute a minimum raise amount 2t for t2 and the expression for β, we obtain:

$A(t) \geq (^1/_2)(t(1 - t)/(1 + t)) + (^1/_2)((\alpha_0 - t)(1+t))(2t) \qquad (\alpha_0 > t)$

Note that we changed the equality to an inequality because in some cases the raise t_2 might be larger than 2t, which would make A(*t*) larger. Simplifying this expression:

$$A(t) \geq \frac{t - t^2 + 2t\alpha_0 - 2t^2}{2(1+t)} \rightarrow \frac{t}{1+t}\left(\frac{1 - 3t + 2\alpha_0}{2}\right)$$

Maximizing this expression, we have:

$A'(t) = (1/(1 + t)^2)(1 - 3t + 2\alpha_0)/2 + (t/(1 + t))(-\,^3/_2)$

$$A'(t) = \frac{1 - 3t + 2\alpha_0 - 3(t + t^2)}{2(1+t)^2}$$

$0 = 1 - 3t + 2\alpha_0 - 3t - t^2$
$3t^2 + 6t + 2\alpha_0 - 1 = 0$
$t^2 + 2t + {}^2/_3\,\alpha_0 - {}^1/_3 = 0$
$(t + 1)2 = {}^2/_3\,\alpha_0 + {}^4/_3$
$t = \sqrt{4/3 + 2/3\alpha_0} - 1$

This / value is therefore a special value; if Y will respond to a check with abet of s_0, then the optimal amount for X to bet with his kings is this value of *t* (related to s_0, of course, because α_0 is a function of s_0). We call this value τ.

We know from previous discussion that X will be indifferent to checking or betting τ for a particular *s*-value because he will play a mixed strategy with aces. (If he does not, Y can exploit him by betting large amounts as we pointed out earlier). We can use this fact to find the proper s_0 value.

$A(0) = A(\tau)$
$(^1/_2)(\alpha_0)(s_0) = (^1/_2)(\tau(1 - \tau)/(1+\tau)) + (^1/_2)((\alpha_0 - \tau)/(1+\tau))(2\tau)$

τ is a function of a_0 and in turn of s_0, so we can solve for s_0. Solving numerically, we obtain $s_0 \approx 0.54224$, $\alpha_0 \approx 0.35159$, and $\tau \approx 0.25209$.

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Recall the previous formula for β(t).

$\beta = (\alpha_0 - t)(1+t)$ $\qquad (\alpha_0 > t)$

It's clear that we cannot raise-bluff a negative percentage of the time. Hence β is zero for values of t greater than α_0. Using this formula for β, we can also find the proper raise-bluffing amount; Y should raise an amount such that X is indifferent to calling with kings (if this is possible - otherwise he should raise-bluff the minimum). If the bet is *t*, and the raise-bluff is to t_2, then we have the following indifference for X betting *t* with a king:

Second player's hand:	**P1 calls with a king**	**P1 folds**
Ace	$-(t_2 - t)$	0
Queen	t_2	$-(1 + t)$

$t_2 - t = \beta(t_2 + 1 + t)$
$(1 - \beta)t_2 = \beta(1 + t) + t$
$(1 - \beta)\, t_2 = \alpha_0 - t + t$
$t_2 = \alpha_0 / (1 - \beta)$

This value t_2 is what we might call the ***theoretical*** value t_2 - the value to which Y would raise if there were no rule limitations on the game requiring that the raise be at minimum twice the original bet. We can immediately see that at bet amounts above α_0 , t_2 is lower than the bet amount; this means that Y need not raise and raise-bluff at all in order to ensure his equity in the game if X bets this value or higher. Also, at *t* values that are very small, Y cannot simply raise to *2t*; by doing so, he loses value against the opposing kings. Instead he raises to a value close to s_0. There is a point close to 0.2001 where the theoretical value is equal to the minimum raise value.

$s(t) = \alpha_0 / (1 - \beta)$ $\qquad (t < 0.2001)$
$s(t) = 2t$ $\qquad (t > 0.2001)$

We now have the outline of the optimal strategies. The only major variables left to find are the checking and betting frequencies of X with aces and kings. What happens here is that X checks aces and kings in a ratio to make Y bet s_0 when X checks, and bets aces and kings in a ratio to make Y indifferent to raise-bluffing the optimal amount. We can then combine these two ratios to find the precise frequencies of each action.

When X checks, he will call $1/(1 + s_0)$ of the time with all his hands to make the opponent indifferent to bluffing with queens. Call k the fraction of checking hands that are kings. If Y bets *q* with an ace when X checks, then X will call with $1/(1 + q)$ of his hands total.

$1 - k$ of those hands will be aces, leaving $1/(1 + q) - 1 + k$ kings out of a total of *k* kings to start.

This amount is $(k - q + kq)/k\,(l + q)$.

Y's equity from betting *q* with an ace when X checks is then:

$(kq - q^2 + kq^2)/k\,(1 + q)$.

Maximizing this expression:

$k(l + q)(k - 2q + 2kq) - k\,(kq - q^2 + kq^2) / k^2(l + q)^2 = 0$

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$k - 2q + 2kq + kq - 2q^2 + 2kq^2 - kq + q^2 - kq^2 = 0$
$k - 2q + 2kq - q^2 + kq^2 = 0$
$k(l + 2q + q^2) = 2q + q^2$

Substituting the known value of s_0, we get:

$k \approx 0.57957$

In a like manner, X has a ratio of aces and kings with which he bets *t*; this ratio is designed to make Y's equity maximum by raise-bluffing the optimal amount. Y gains $1 + \tau$ from successfully bluffing, but loses his raise amount *q* when called. X, however, will not call with his kings! He can simply fold all his kings to the raise; Y can no longer exploit him by raising a smaller amount because the raise amount must be at least 2*t*. He will call enough to make Y indifferent to raise-bluffing, or $(1 + \tau)/(1 + \tau + q)$ of the time.

So his ratio of kings to aces for betting is $q/(l + \tau + q) \approx 0.28707$.

Utilizing these two ratios, we can then find the full strategy for aces and kings. Letting c be the total amount of hands checked and b the total amount of hands bet, we have:

$0.28707b + 0.57957c = 1$
$0.71293b + 0.42043c = 1$
$b \approx 1.45595$
$c \approx 0.55405$
$A_k = 61.213\%$
$A_b = 38.787\%$
$K_k = 84.382\%$
$K_b = 15.618\%$

And with that, we have the optimal strategy to the game.

The value of this game to Y is approximately 0.02682 units per play.

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